CHAPTER 1

What Is Information?

It is always tricky to define fundamental concepts. Since they are the basis for everything that follows, they themselves seem to lack any basis of their own. Physicists have a hard time defining matter and energy, biologists have a hard time defining life, and philosophers have a hard time defining reality.

Information is increasingly seen by many philosophers and biologists, and even by some physicists, as the most basic building block of reality, more elementary than matter and energy.[1] No wonder that there are many disputes about how to define information, and how it is related to the evolution of life or to basic ideas in physics such as entropy, the laws of thermodynamics and the quantum uncertainty principle.[2] This book will make no attempt to resolve – or even explain – these disputes, nor will it offer a universal definition of information applicable to physics, biology and all other fields of knowledge. Since it is a work of history, which studies the past and future development of human societies, it will focus on the definition and role of information in history.

In everyday usage, ‘information’ is associated with human-made symbols like spoken or written words. Consider, for example, the story of Cher Ami and the Lost Battalion. In October 1918, when the American Expeditionary Forces were fighting to liberate northern France from the Germans, a battalion of more than five hundred American soldiers was trapped behind enemy lines. American artillery, which was trying to provide them with cover fire, misidentified their location and dropped the barrage directly on them. The battalion’s commander, Major Charles Whittlesey, urgently needed to inform headquarters of his true location, but no runner could